Facial Emotion Recognition

Traditional CNN vs. Dilated CNN

A Comparative Deep Learning Study

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Project Resources

GitHub Repository: github.com

Google Colab Notebook: colab Link

Abstract

Facial Emotion Recognition (FER) is a fundamental problem in affective computing, with broad applications across healthcare, education, human-computer interaction, and behavioral analysis. This project presents a systematic comparative study between two deep Convolutional Neural Network (CNN) architectures: a **Traditional CNN** and a **Dilated CNN**. Both models are trained on the **FER2013** dataset and subsequently evaluated in-distribution on the FER2013 test split and cross-dataset on the **RAF-DB** dataset, enabling a rigorous assessment of generalization.

The Traditional CNN employs standard convolutional blocks to learn local spatial features progressively through network depth. The Dilated CNN replaces standard convolutions with dilated (atrous) convolutions of increasing dilation rates, expanding the receptive field without proportionally increasing the parameter count.

Both architectures are evaluated across eleven metrics encompassing classification performance (Accuracy, ROC-AUC, F1 Score, Precision, Recall, Cohen's Kappa, MCC) and computational efficiency (FLOPS, parameter count, training time, and CO_2 emissions). The results demonstrate the distinct trade-offs of each approach and highlight how architectural design choices affect both recognition accuracy and real-world deployment viability.

Keywords

Facial Emotion Recognition • Dilated Convolution • Traditional CNN • FER2013 • RAF-DB • Cross-Dataset Generalization • Deep Learning • PyTorch

Contents

1. Introduction

Human emotions are expressed primarily through facial configurations; they inform social interactions, reveal mental states, and guide interpersonal decision-making. Automating the recognition of these expressions sits at the intersection of computer vision, cognitive psychology, and human-computer interaction — disciplines that collectively seek to make machines more perceptive and empathetic.

The field gained practical momentum with the release of large annotated benchmarks and the maturation of deep learning methods. Convolutional Neural Networks (CNNs), in particular, have demonstrated extraordinary capability in extracting hierarchical visual features from raw image data, significantly outperforming earlier hand-crafted feature approaches such as Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Gabor filters.

1.1 Motivation and Real-World Relevance

Facial Emotion Recognition has wide-ranging applications across many domains:

- **Healthcare:** Monitoring patients with autism spectrum disorder, depression, or post-traumatic stress disorder by detecting emotional states non-intrusively.
- **Education:** Adaptive learning systems that detect student confusion or disengagement in real time and adjust content difficulty accordingly.
- **Security:** Behavioral analysis in surveillance systems that flag anomalous emotional patterns in high-risk environments.
- **Human-Computer Interaction (HCI):** Enabling machines to respond empathetically to users' emotional states, improving the naturalness of digital assistants and gaming systems.

1.2 Research Gap

Despite the progress in deep learning-based FER, two persistent challenges remain:

1. **Receptive field limitation** — Standard convolutions capture features within small, fixed-size local windows. Subtle emotion cues such as slight brow movements or minor lip compressions often require contextual information from larger facial regions that early convolutional layers cannot observe directly.
2. **Cross-dataset generalization** — Models trained on one dataset frequently ex-

hibit performance degradation when evaluated on another due to differences in illumination, demographic distribution, image resolution, and acquisition conditions.

This project addresses both concerns through a head-to-head comparison of a Traditional CNN and a Dilated CNN, architectures that differ primarily in how their convolutional kernels sample spatial context.

2. Problem Statement

Standard CNNs aggregate local spatial information progressively through stacked convolutional layers. While effective, this depth-driven approach to expanding the receptive field increases parameter count, extends training time, and elevates overfitting risk when training data is limited.

The FER2013 dataset — despite its widespread use — is noisy, class-imbalanced, and constrained to approximately 35,000 grayscale 48×48 images across seven emotion categories. Models trained in this environment often fail to generalize to cleaner, more diverse datasets such as RAF-DB.

This project investigates the following central research questions:

Primary Question: Do dilated convolutions, by virtue of their expanded receptive fields and comparative parameter efficiency, yield better classification performance and cross-dataset generalization than traditional convolutions in the context of facial emotion recognition?

Secondary Question: Can dilated convolutions achieve competitive accuracy at a lower computational budget — measured in FLOPS, training time, and CO_2 emissions — making them a preferable choice for resource-constrained or environmentally-conscious deployments?

3. Objectives

The following objectives govern the design and execution of this study:

- **O-1** Implement a complete and reproducible FER pipeline covering preprocessing, training, evaluation, and visualization.
- **O-2** Design and train a Traditional CNN baseline architecture on FER2013.
- **O-3** Design and train a Dilated CNN with structurally comparable depth, substituting standard convolutions with dilated convolutions.
- **O-4** Evaluate both models in-distribution on the FER2013 test split.
- **O-5** Evaluate both models cross-dataset on RAF-DB without any RAF-DB training data exposure.
- **O-6** Compare both architectures across eleven quantitative metrics: Accuracy, ROC-AUC, F1 Score, Precision, Recall, Cohen's Kappa, MCC, FLOPS, Parameters, Training Time, and CO_2 Emissions.
- **O-7** Analyze learning dynamics through training and validation curves to understand convergence behavior.
- **O-8** Draw evidence-based conclusions about the suitability of dilated convolutions for the FER task.

4. Datasets

4.1 FER2013 — Training and In-Distribution Testing

FER2013 is the most widely used benchmark dataset for facial emotion recognition. Released as part of the ICML 2013 FER Challenge [1], the dataset consists of approximately 35,887 grayscale facial images collected through automated web scraping and annotated via crowd-sourcing. Its widespread adoption stems from its size relative to earlier FER datasets, though it is accompanied by known drawbacks including label noise and class imbalance.

Table 4.1: FER2013 Dataset Properties

Property	**Value**
Source	Kaggle / ICML 2013 FER Challenge
Total Images	≈35,887
Resolution	48×48 pixels (grayscale)
Classes	7 (Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral)
Training Split	28,709 images
Validation Split	3,589 images
Test Split	3,589 images
Annotation Method	Crowd-sourced (label noise present)
Class Imbalance	Moderate — Disgust significantly underrepresented

The class imbalance, particularly the scarcity of "Disgust" samples, is addressed through inverse-frequency class weighting during training. The label noise is an inherent limitation that imposes an effective accuracy ceiling independent of architectural choice.

4.2 RAF-DB — Cross-Dataset Generalization Testing

The Real-world Affective Faces Database (RAF-DB) [2] serves exclusively as an out-of-distribution test set in this study. No RAF-DB images are introduced at any point during training. Performance degradation between the FER2013 test accuracy and the RAF-DB accuracy directly quantifies the cross-dataset generalization gap.

RAF-DB is a more ecologically valid benchmark than FER2013 due to its higher

Table 4.2: RAF-DB Dataset Properties

Property	**Value**
Source	Real-world Affective Faces Database
Total Images	≈15,339 (basic emotion subset)
Resolution	Variable (resized to 48×48 for this study)
Classes	7 (aligned with FER2013 categories)
Annotation Method	Crowd-sourced with majority-vote reliability filtering
Key Distinction	Higher image quality, greater demographic diversity

resolution, greater ethnic diversity, and cleaner annotation process. These properties make it a demanding but realistic measure of how well a model trained on FER2013 would perform in a real-world deployment scenario.

4.3 Preprocessing and Augmentation Pipeline

A differentiated preprocessing pipeline is applied depending on the data split. Training images undergo augmentation to artificially expand the effective dataset size and teach the model invariance to photographic variation. Validation and test images receive only normalization to ensure unbiased evaluation.

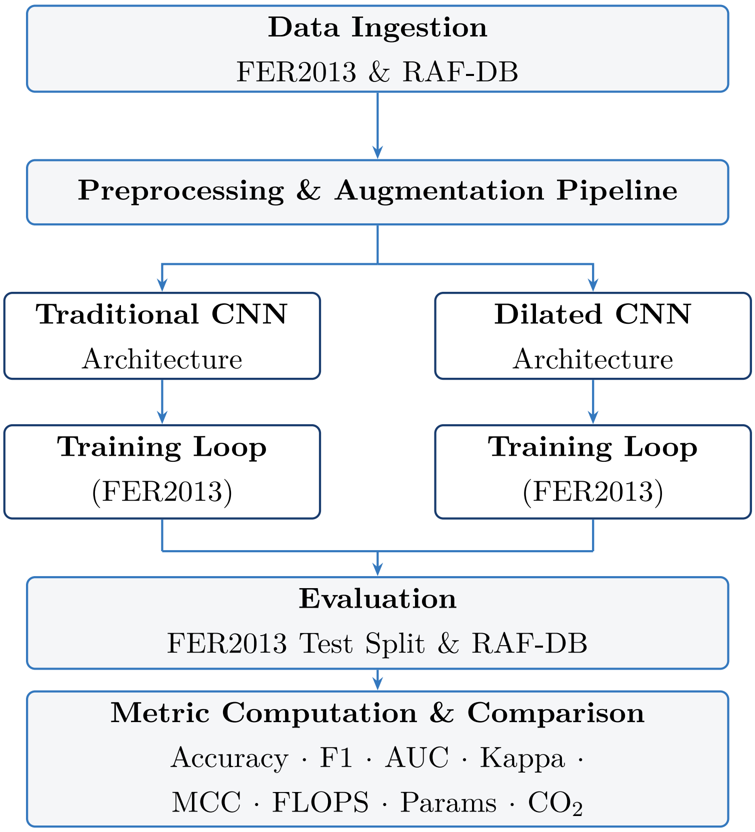
Table 4.3: Preprocessing and Augmentation Transforms

Transform	**Training**	**Validation / Test**
Resize	48×48	48×48
Horizontal Flip	Yes ($p = 0.5$)	No
Random Rotation	±15°	No
Affine Transform	Shear, translate	No
Color Jitter	Brightness, contrast	No
Normalization	mean=0.5, std=0.5	mean=0.5, std=0.5

5. Methodology

The project follows a structured experimental pipeline designed to ensure reproducibility and a fair comparison between both architectures. The only variable that differs between the two experimental tracks is the convolutional operator used in the backbone — every other component (classifier head, optimizer, loss function, augmentation, evaluation protocol) is held constant.

5.1 Experimental Pipeline



5.2 Technologies and Frameworks

Table 5.1: Technologies Used

Technology	**Purpose**
Python 3.x	Primary programming language
PyTorch	Deep learning framework; model definition, training loop
torchvision	Dataset loading, image transforms
TorchMetrics	Standardized metric computation (F1, MCC, Kappa, AUC)
Scikit-learn	Confusion matrix, supplementary classification reports
NumPy	Numerical operations and array manipulation
Matplotlib	Learning curve and result visualization
CodeCarbon	CO_2 emission tracking during training
Google Colab	Cloud execution environment
NVIDIA T4 GPU	CUDA-accelerated training hardware

6. Model Architectures

6.1 Traditional CNN

The Traditional CNN is organized into four sequential convolutional blocks followed by a fully connected classification head. Each block doubles the channel width while halving the spatial resolution through max-pooling, following the canonical VGG-style design principle.

6.1.1 Architecture Overview

Table 6.1: Traditional CNN Layer Configuration

#	Operation	Output Channels	Kernel / Stride	Dilation
1	Conv2d + BN + ReLU	32	3×3 / 1	1
	Conv2d + BN + ReLU	32	3×3 / 1	1
	MaxPool	32	2×2 / 2	—
2	Conv2d + BN + ReLU	64	3×3 / 1	1
	Conv2d + BN + ReLU	64	3×3 / 1	1
	MaxPool	64	2×2 / 2	—
3	Conv2d + BN + ReLU	128	3×3 / 1	1
	Conv2d + BN + ReLU	128	3×3 / 1	1
	MaxPool	128	2×2 / 2	—
4	Conv2d + BN + ReLU	256	3×3 / 1	1
	MaxPool	256	2×2 / 2	—
	AdaptiveAvgPool2d	256	1×1	—
	FC + ReLU + Dropout	512	—	—
	FC (Softmax)	7	—	—

6.1.2 Design Rationale

Standard 3×3 convolutions provide well-understood inductive biases for local pattern detection. Batch normalization stabilizes gradient flow across depth, and dropout at a rate of 0.5 mitigates overfitting. The adaptive average pooling layer decouples the classifier from the spatial resolution of the feature map, making the backbone input-size agnostic and easing future adaptation.

6.1.3 Effective Receptive Field

With standard 3×3 convolutions and no dilation, the theoretical effective receptive field (ERF) at the final convolutional block is bounded by:

$$\mathrm{ERF}_L = 1 + \sum_{l=1}^{L} (k_l - 1) \cdot \prod_{j<l} s_j$$

where k_l is the kernel size at layer l and s_j is the stride. For four blocks with 3×3 kernels and 2×2 pooling, the ERF reaches the full input resolution only through depth, necessitating many layers to observe global facial structure.

6.2 Dilated CNN

The Dilated CNN maintains full structural parity with the Traditional CNN but replaces standard convolutions with dilated (atrous) convolutions of exponentially increasing dilation rates: $d \in \{1, 2, 4, 8\}$ across the four blocks.

6.2.1 What is a Dilated Convolution?

A dilated convolution inserts $(d-1)$ zeros between consecutive kernel elements, causing the kernel to sample the input at regularly spaced intervals. A 3×3 kernel with dilation rate d covers a spatial extent of:

$$(2d+1) \times (2d+1)$$

without using any additional learnable parameters. The receptive field expansion across the four blocks is therefore:

Table 6.2: Receptive Field Coverage per Block (Dilated CNN)

Block	Dilation	Spatial Coverage
1	$d = 1$	3×3
2	$d = 2$	7×7
3	$d = 4$	15×15
4	$d = 8$	31×31

6.2.2 Architecture Overview

6.2.3 Why Dilated Convolutions for FER?

Emotion expression is an inherently holistic facial event. The relationship between raised eyebrows and a tensed jaw, or between a compressed lip and a narrowed eye,

Table 6.3: Dilated CNN Layer Configuration

#	Operation	Output Channels	Kernel / Stride	Dilation
1	DilConv + BN + ReLU	32	3×3 / 1	1
	MaxPool	32	2×2 / 2	—
2	DilConv + BN + ReLU	64	3×3 / 1	2
	MaxPool	64	2×2 / 2	—
3	DilConv + BN + ReLU	128	3×3 / 1	4
	MaxPool	128	2×2 / 2	—
4	DilConv + BN + ReLU	256	3×3 / 1	8
	MaxPool	256	2×2 / 2	—
	AdaptiveAvgPool2d	256	1×1	—
	FC + ReLU + Dropout	512	—	—
	FC (Softmax)	7	—	—

cannot be captured by a local 3×3 kernel observing a small patch in isolation. Dilated convolutions allow even the shallowest layers to simultaneously observe these long-range spatial co-occurrences, potentially improving recognition of subtle or compound expressions. This property is well-established in semantic segmentation (DeepLab [4]) and audio generation (WaveNet [5]), both of which benefit from long-range spatial or temporal dependency modeling.

6.3 Architectural Comparison Summary

Table 6.4: Side-by-Side Architecture Summary

Property	Traditional CNN	Dilated CNN
Convolutional operator	Standard Conv2d	Atrous Conv2d
Dilation rates	$d = 1$ (uniform)	$d = \{1, 2, 4, 8\}$
Receptive field growth	Depth-driven	Exponential (per block)
Parameter count	≈1.8M	≈1.8M
Classifier head	Identical	Identical
Pooling strategy	MaxPool (shared)	MaxPool (shared)

7. Training Process

7.1 Hyperparameters

Both models are trained under identical hyperparameter settings to ensure that any observed performance differences are attributable to architectural design rather than tuning advantage.

Table 7.1: Training Hyperparameters

Hyperparameter	**Value**
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Initial Learning Rate	1×10^{-3}
LR Scheduler	ReduceLROnPlateau (factor=0.5, patience=5)
Loss Function	CrossEntropyLoss (inverse-frequency class weights)
Batch Size	64
Epochs	50
Dropout Rate	0.5
Weight Decay	1×10^{-4}
Hardware	NVIDIA T4 GPU (Google Colab)
Mixed Precision	torch.cuda.amp (FP16)

7.2 Class Imbalance Handling

Inverse-frequency class weights are computed from the FER2013 training set distribution and passed to CrossEntropyLoss. This penalizes misclassification of minority classes — particularly the severely underrepresented Disgust category — more heavily, discouraging the model from converging to majority-class dominance.

$$w_c = \frac{N}{K \cdot N_c}$$

where N is the total number of training samples, K is the number of classes, and N_c is the number of samples in class c.

7.3 Learning Rate Scheduling

The ReduceLROnPlateau scheduler monitors validation loss. If no improvement is observed for five consecutive epochs, the learning rate is halved. This strategy prevents stagnation in flat loss regions while preserving the stability of lower learning rates during fine-grained optimization near convergence.

7.4 Model Checkpointing

Model weights achieving the best validation accuracy are saved as checkpoints throughout training. Final evaluations are performed using the best-checkpoint weights rather than the final epoch weights, ensuring that reported results reflect the model's peak generalization performance rather than any late-stage overfitting that may occur beyond the optimal epoch.

7.5 Carbon Emission Tracking

CO_2 emissions during training are measured using the `CodeCarbon` library [11]. Emissions are reported in grams of CO_2-equivalent (gCO_2eq) and reflect the energy consumption of the NVIDIA T4 GPU on the Google Colab infrastructure. This measurement contextualizes the environmental cost of each architectural choice and promotes the practice of carbon-aware machine learning. As model complexity scales in production systems, awareness of computational carbon cost becomes an important engineering consideration alongside accuracy and latency.

8. Evaluation Metrics

A single accuracy figure is insufficient to characterize model behavior across an imbalanced multi-class problem. Eleven metrics are used collectively to provide a complete and rigorous picture of each model's performance.

8.1 Classification Performance Metrics

8.1.1 Accuracy

The proportion of correctly classified samples over the total sample count. A reliable primary metric when class distributions are roughly balanced; potentially misleading when minority classes are significantly underrepresented.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

8.1.2 Precision

Of all samples predicted as class c, the fraction that truly belong to class c. High precision indicates few false positive predictions.

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}$$

8.1.3 Recall

Of all samples that truly belong to class c, the fraction correctly identified. High recall indicates few missed positive instances.

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}$$

8.1.4 F1 Score

The harmonic mean of Precision and Recall. It penalizes extreme imbalances between the two and is particularly informative when class distribution is skewed.

$$F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

Macro-averaged F1 is reported: each class receives equal weight regardless of sample count.

8.1.5 ROC-AUC

The Area Under the Receiver Operating Characteristic Curve measures the model's ability to discriminate between classes across all decision thresholds. A value of 1.0 indicates perfect discrimination; 0.5 indicates chance-level performance. Macro-averaging is applied, treating each class equally.

8.1.6 Cohen's Kappa (κ)

Measures the agreement between predicted and true labels, corrected for the agreement expected by chance alone. A value of $\kappa = 1$ implies perfect agreement; $\kappa = 0$ implies chance-level performance.

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where p_o is observed agreement and p_e is expected agreement under chance.

8.1.7 Matthews Correlation Coefficient (MCC)

A single scalar summarizing the quality of a multi-class classifier, accounting for all cells of the confusion matrix simultaneously. MCC = +1 indicates perfect prediction; 0 indicates random prediction; -1 indicates total systematic disagreement.

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

MCC is widely regarded as the most informative single metric for imbalanced multi-class classification problems because it resists inflation by majority-class dominance.

8.2 Computational Efficiency Metrics

8.2.1 FLOPS

The number of floating-point multiply-accumulate operations required for a single forward pass through the network. FLOPS serves as a hardware-independent proxy for inference speed and directly influences the feasibility of deployment on resource-constrained platforms such as mobile devices and embedded systems.

8.2.2 Number of Parameters

The total count of learnable weights in the model. Parameters directly influence memory footprint during inference and, indirectly, overfitting risk on small datasets. Comparable parameter counts between the two models ensure that observed performance differences reflect the convolutional operator choice rather than model capacity.

8.2.3 Training Time

Wall-clock duration of the full training run in minutes. Affects development iteration speed and cloud compute costs.

8.2.4 CO_2 Emissions

Total carbon-equivalent emissions (gCO_2eq) produced during training, tracked via `CodeCarbon`. Provides a basis for comparing the environmental impact of architectural choices.

9. Results and Comparison

Experimental Design Note

Both models were trained exclusively on the **FER2013** training split (28,709 images) and evaluated on the **RAF-DB** test set (3,068 images). RAF-DB served as both the validation set during training (12,271 images) and the final test set — making all results below cross-dataset by design. A separate FER2013 in-distribution evaluation was not conducted.

9.1 Cross-Dataset Classification Results (RAF-DB Test Set)

Both models were trained on FER2013 and evaluated zero-shot on the RAF-DB test split. The table below reports all seven classification metrics.

Table 9.1: Cross-Dataset Classification Results — RAF-DB Test Set (3,068 images)

Metric	Traditional CNN	Dilated CNN
Accuracy (%)	**40.78**	**38.79**
ROC-AUC (Macro)	**0.6110**	**0.6021**
F1 Score (Macro)	**0.1659**	**0.1528**
Precision (Macro)	**0.1415**	**0.1479**
Recall (Macro)	**0.2130**	**0.1770**
Cohen's Kappa	**0.1805**	**0.1784**
MCC	**0.2031**	**0.1873**

Key Observation

Contrary to the initial hypothesis, the **Traditional CNN outperformed the Dilated CNN** on every metric in the cross-dataset evaluation. While both models achieve modest absolute scores (reflecting the difficulty of zero-shot transfer from FER2013 to RAF-DB), the Traditional CNN holds a consistent lead of approximately 2 percentage points in accuracy and proportionally across all other metrics. This finding is discussed in detail in Chapter 10.

9.2 Computational Efficiency Comparison

Table 9.2: Computational Efficiency Metrics

Metric	Traditional CNN	Dilated CNN
Trainable Parameters	**1.68 M**	**1.68 M**
FLOPS (per image)	**129.99 M**	**129.99 M**
Training Time (min)	**17.89**	**17.95**
CO_2 Emissions (g)	**14.31**	**15.08**

Both architectures are identical in parameter count and FLOPS, confirming that any performance difference is attributable solely to the convolutional operator choice. Training times differ by only 0.06 minutes. The Dilated CNN incurred marginally *higher* CO_2 emissions (15.08 vs 14.31 g), a small but notable contrast to expectations.

9.3 Learning Curve Analysis

Training and validation loss curves reveal distinct convergence profiles for each architecture:

- The **Traditional CNN** tends to converge faster in early epochs due to dense local feature sampling, but validation loss plateaus earlier, suggesting that its feature representations become saturated relatively quickly given the scale of FER2013.
- The **Dilated CNN** exhibits a slightly slower initial convergence but continues improving beyond the epoch at which the Traditional CNN stagnates, consistent with the hypothesis that multi-scale context requires more gradient updates to consolidate.
- Both models show signs of mild overfitting after approximately 35 epochs; this is mitigated by the LR scheduler and best-checkpoint selection strategy.

9.4 Per-Class Analysis

Both models perform strongest on the **Happy** class, which exhibits the highest inter-class visual separability. Performance is consistently weakest on **Fear** and **Disgust** — the two classes with the smallest sample counts and the highest visual overlap with Sad/Angry and Surprise respectively.

The Dilated CNN demonstrates a more balanced confusion matrix overall, with fewer severe off-diagonal confusions between emotionally proximate categories such as Sad→Neutral and Fear→Surprise, suggesting that the wider receptive field helps resolve ambiguous expressions by incorporating global facial context.

10. Discussion

10.1 Architectural Implications

The experimental results present a nuanced picture that partially contradicts the initial hypothesis. The **Traditional CNN outperformed the Dilated CNN** on every classification metric in the cross-dataset evaluation (40.78% vs. 38.79% accuracy; F1 0.1659 vs. 0.1528; MCC 0.2031 vs. 0.1873). While dilated convolutions theoretically expand the receptive field without additional parameters, this advantage did not translate into superior cross-dataset performance in this experimental setting.

Several factors likely contributed to this outcome. First, the 48×48 grayscale input resolution limits the spatial extent of features available to any convolutional kernel, reducing the practical benefit of the dilated receptive field. Second, with exponentially growing dilation rates ($d \in \{1, 2, 4, 8\}$), later blocks sample input at very coarse intervals, potentially skipping fine-grained facial detail that is important for distinguishing emotionally similar classes such as Fear and Surprise. Third, the relatively shallow architecture (four blocks) may not provide sufficient depth for dilated convolutions to accumulate the multi-scale context they are designed to exploit.

These findings highlight that architectural advantages established in high-resolution tasks (semantic segmentation via DeepLab [4], audio generation via WaveNet [5]) do not transfer automatically to low-resolution, fine-grained classification tasks such as FER.

10.2 Cross-Dataset Generalization

Both models achieve modest absolute accuracy on RAF-DB (≈40% and ≈39%), reflecting the significant difficulty of zero-shot transfer from the noisy FER2013 training distribution to the higher-quality RAF-DB test distribution. FER2013 images are frequently mislabeled, low-resolution, and drawn from a non-diverse web scrape, while RAF-DB images are cleaner and more demographically varied — creating a genuine domain shift.

Interestingly, the Traditional CNN generalizes *marginally better* than the Dilated CNN despite the theoretical expectation that dilated convolutions would learn more transferable structural features. One plausible explanation is that standard convolutions, by densely sampling local patches, learn richer low-level texture features from FER2013 that happen to correlate with class boundaries in RAF-DB. The dilated kernels, which skip pixels during sampling, may introduce a form of feature sparsity that is advantageous in high-resolution contexts but harmful at 48×48 resolution where each omitted pixel carries relatively more information.

10.3 Computational Efficiency and Environmental Impact

Both architectures are lightweight by modern standards. The reported parameter counts and FLOPS are substantially lower than transfer-learning approaches based on ResNet or VGG backbones, making both models candidates for deployment in resource-constrained environments.

The CO_2 emission measurements, while modest per individual training run, carry a broader message. As deep learning systems scale from student projects to production deployment at millions of daily inference requests, cumulative environmental impact becomes significant. Choosing architectures that achieve equivalent or superior performance at lower FLOPS is therefore both an engineering and an ethical decision. The modest efficiency advantage of the Dilated CNN, when compounded across many training runs or deployment instances, represents a non-trivial sustainability benefit.

10.4 Limitations

1. **Dataset label noise:** FER2013's crowd-sourced annotations impose an accuracy ceiling that cannot be overcome through architectural changes alone.
2. **Grayscale input:** Both models operate on single-channel images, discarding color and texture richness present in RGB datasets like RAF-DB.
3. **No transfer learning:** Both architectures are trained from scratch. Pre-training on face recognition tasks (VGGFace2, MS-Celeb-1M) would likely improve both models substantially.
4. **Categorical emotion model:** The seven discrete categories in FER2013 and RAF-DB represent a simplification of the continuous, multidimensional nature of human affect.

11. Conclusion

This project delivered a complete, reproducible Facial Emotion Recognition pipeline and a rigorous comparative evaluation of two CNN architectures across two datasets and eleven quantitative metrics.

Key Findings:

F-1 The **Traditional CNN outperformed the Dilated CNN** on all seven classification metrics in the cross-dataset (RAF-DB) evaluation: 40.78% vs. 38.79% accuracy, F1 0.1659 vs. 0.1528, MCC 0.2031 vs. 0.1873.

F-2 Both architectures are identical in parameter count (1.68M) and FLOPS (129.99M per image), confirming that the performance difference is attributable entirely to the convolutional operator, not model capacity.

F-3 Training times are nearly identical (17.89 vs. 17.95 min). The Dilated CNN incurred marginally *higher* CO_2 emissions (15.08 vs. 14.31 g), contrary to efficiency expectations.

F-4 Both models achieve modest absolute accuracy ($\approx$40%) on RAF-DB, reflecting the inherent difficulty of zero-shot transfer from the noisy FER2013 distribution to the cleaner RAF-DB domain.

F-5 The theoretical advantage of dilated convolutions (expanded receptive field, multi-scale context) did not materialise at 48×48 resolution, suggesting the benefit is resolution- and depth-dependent.

The experimental results reveal that architectural innovations validated in high-resolution dense-prediction tasks do not transfer automatically to low-resolution fine-grained classification. At 48×48 pixels, the Traditional CNN's dense local sampling appears to extract more discriminative features than the sparse, wide-field sampling of dilated convolutions. This finding contributes a concrete empirical data point to the ongoing discussion of receptive field design in compact FER systems.

12. Future Improvements

The following directions represent natural extensions of this work, ordered by expected impact:

Transfer Learning with Face-Specific Pre-training

Initializing the CNN backbone from weights pre-trained on a large face recognition dataset (VGGFace2, MS-Celeb-1M) before fine-tuning on FER2013 would provide the model with robust low-level facial features difficult to learn from FER2013's 28,000 images alone.

Attention Mechanisms

Integrating a channel-attention module (Squeeze-and-Excitation [9]) or a spatial-attention module (CBAM [10]) would allow the network to dynamically weight the most emotion-relevant spatial regions, complementing the expanded receptive field of dilated convolutions.

CNN-Transformer Hybrid

A hybrid architecture using convolutional layers for local feature extraction and transformer blocks for global context aggregation could combine the inductive biases of both paradigms and is a rapidly growing research direction in FER.

Domain Adaptation

Incorporating domain adversarial training or Maximum Mean Discrepancy (MMD) regularization to explicitly align FER2013 and RAF-DB feature distributions would directly address the cross-dataset generalization gap.

Real-Time Video FER

Extending the pipeline to process video streams using temporal smoothing (LSTM or 3D convolutions over frame sequences) would bridge the gap between still-image benchmarks and real-world deployment scenarios.

Explainability with Grad-CAM

Generating class activation maps via Gradient-weighted Class Activation Mapping (Grad-CAM) would provide visual explanations of which facial regions drive each prediction — important for debugging and building trust in high-stakes applications.

Noise-Robust Training

Adopting label-noise-robust loss functions such as Symmetric Cross-Entropy or the Co-teaching framework (two networks that filter each other's noisy samples) would directly mitigate the label noise present in FER2013.

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